What I Did, What I Found, What I Need

The printout: the project from zero, the evidence, and four decisions

Dheirav Prakash

What I did, what I found, and what I need from you

Where this started

This began as the course project for Responsible AI: measure and mitigate bias in an income classifier on the standard UCI Adult benchmark. That part is complete and submitted. While doing it I noticed something the assignment did not ask about, and I kept pulling on it after the deliverable was done. Everything in this document is individual work beyond the course submission, and all of it is committed in a public repository where every number below is re-derived from stored results by automated checks.

The observation that started it

A fairness constraint asks a model to approve two groups at similar rates, but it does not say how the gap should close. The model can approve more people from the group that was behind, or approve fewer from the group that was ahead. Both count as success, and the fairness metric that certifies the fix comes out identical in both cases. So a team imposing a constraint cannot tell, from its own reporting, whether it is about to extend opportunities or withdraw them.

On Adult, every mitigation method I tested closed the gap the second way. The total pool of approvals shrank by 8 to 22 percent depending on the method, and when I counted people instead of rates, 66 to 74 percent of everyone whose decision changed was a person losing an approval. The metric reported success identically throughout, which meant the interesting question was not whether this happens but when.

The finding

I measured what decides the direction, across what grew to 67 populations from 8 data sources in 5 countries: US census data across forty states and multiple years, real mortgage decisions from eight state markets, criminal justice, education, the Dutch census, Taiwanese credit, and census extracts from Brazil and Mexico.

Within a population, the baseline approval rate read against that population's own switch point predicts whether the constraint will grow or shrink the pool. Stingy systems lose approvals when you impose parity, while generous systems gain them, and the switch between the two sits at a measurable crossover. On mortgage data, where approval rates run above 0.8, the same constraint that shrinks Adult's pool by a fifth grows the pool instead, in 7 of the 8 markets tested. The claim covers direction only: the size of the effect is ordered within a population but does not transfer between populations, and a model of size that I registered in advance lost to predicting zero, so the paper does not claim it.

There is a transported shortcut, "below roughly 0.54 the fix takes, above it gives", and it works often but not always. It called 9 of 10 never-measured states correctly in a prediction committed to the repository before the data existed, but it managed only 5 of 10 on a later set of larger states, and located crossovers now span 0.28 to 0.85. So the honest form of the finding is: measure your own crossover with the audit procedure I built, and treat 0.54 as a starting guess rather than a law.

How I made sure this is real

Correlations found after the fact are weak evidence for a claim of the form "you can know in advance", so the project runs on pre-registered predictions. A prediction, its pass mark, and the naive alternative it must beat are committed to the repository before the data exists, and the recent commitments are also timestamped into the Bitcoin blockchain so the ordering does not depend on trusting my clock.

Sixteen such tests now stand. Ten failed or were refused, and every failure is published uncorrected, because each one removed a claim stronger than the one that survives. The ones that matter most:

Test	Outcome	What it settled
Sealed direction rule, refined	4 of 8, fail	a clause I added from early data was wrong, and is withdrawn
Re-seal of the simple rule	9 of 10, holds	the rule predicts unseen states from the rate alone
Sealed magnitude model	fail	effect size cannot be predicted; direction only
Post-processing transfer	fail	the rule vanishes when the fix is applied after training
Regime deconfounding	method reading	see the correction below
Six-market lending seal	protocol fail, direction holds	my sweep procedure is unreliable on mortgages, while the direction held on 7 of 8 markets
Brazil and Mexico, sex arms	refused	Brazil has almost no conditional sex gap, so my own gates declined to answer
Brazil race arms, screen-gated	rate wins	see below: the biggest open question, answered

Two results from the final campaigns deserve their own lines. First, the one standing worry about the 9-of-10 was that those arms reached their rates through an income cutoff, so a rule reading only the cutoff would have scored identically, which meant the evidence could not separate "the approval rate predicts" from "the label's rarity predicts". A powered test on Brazil's racial gap separated them: the cutoff-only reading scored 5 of 10 while the rate scored 8, so it is the rate. Second, the 0.54 value itself earned no special status in that test, which is consistent with everything above: the portable thing is the procedure, not the number.

What I got wrong, and what correcting it taught

Two of the project's biggest findings came from being wrong in public.

For two review cycles I claimed the rule works only when the model cannot see the protected attribute, and I read that as matching the boundary a related theory paper draws. A reviewer pointed out my evidence changed two things at once, so I sealed the missing experiment with both possible readings' pass marks committed in advance. The result went against my own interpretation: the same method with the attribute visible keeps the relationship fully intact, so the boundary is the kind of method, not blindness, and the rule applies more widely than I had claimed.

Separately, four 2022 states appeared to invert the whole relationship. The cause turned out to be that the "earns over $50,000" label had slid down the income distribution, and a strict inflation correction was not enough because real incomes grew past it. What restores normal behaviour is matching the label's position in the distribution, which is a general lesson: outcome definitions anchored to fixed dollar amounts are a different task every year without anyone noticing.

One more finding worth naming because it is fully mine: at severe operating points the constraint achieves equality by discarding approvals at random, so a person's own score stops mattering, under a certificate that looks like an informed adjustment. I verified this inside the fitted models, showed it does not happen at any normal operating point tested, and showed no alternative of the same kind exists where it does happen. A citation-graph audit found this observation nowhere in the literature.

How this relates to the existing theory

A March 2026 theory paper proved that in this setting the direction can go either way depending on the data. That paper contains no experiments, never mentions an approval rate, and states its conditions over quantities that could not be evaluated on any of 26 real datasets under either of two estimators I tried. My contribution relative to it: the computable predictor it lacks, the validation record, the boundary map, and one correction to where its regime line bites in practice, while its own group-level prediction held in every one of the 45 attribute-aware populations I measured. The approval rate tracks their theoretical quantity at a correlation of 0.85 across 150 populations, so the number I use appears to be the practical stand-in for the quantity their theorem is written over.

Where it stands

The paper is 18 pages in IEEE format with a written plan to compress to about 10 once a venue is chosen. The record behind it: 68 numbered research documents, 55 automated checks that fail if any documented number disagrees with stored results, and every pre-registration verifiable from the public repository.

The decisions I need from you

1. **Venue and page limit.** The work is shaped like FAccT or AIES. Your answer decides the compression depth, and the plan is already written.
1. **Does the empirical half stand as a paper, given the theory paper exists?** My position: their theorem is not computable on data and mine is the instrument that reads it, which is a recognized kind of contribution. This is the call I most need from you.
1. **The title.** The current title claims a scope my own sealed test dismantled. Two bounded replacements are drafted, and I would like your pick.
1. **Contacting the theory paper's authors.** My results confirm their group-level prediction everywhere, correct their regime boundary, and hand them an open theory problem. Standard practice is a short collegial email once a preprint exists. I want your read on whether and when, and whether you want to be on it.